



Logic in AI

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1 INTRODUCTION

There are two main goals for this course:

- Introducing a notion of deduction
- Introducing Curry Howard Isomorphism, that connect the notion of computation with the notion of deduction. A proof is just a mathematical object that can be manipulated and transformed.

Definition: Logical System

A logical system has at least three components:

- Syntax (an alphabet, a grammar, a set of well-formed formulas)
- Semantics (in order to link a meaning to the syntax)
- Deductive system (a set of rules to derive new formulas from the existing ones)

Perhaps, in the past you have encountered a logical system like Natural Deduction, tableaux, or sequent calculus.

1.1 AREAS OF INTEREST

There are several applications of logic in AI, took directly from SOTA (state of art) papers:

- **“Theoretical” Computer Science:** automata, formal languages, computability or recursion theory, complexity theory, etc.
- **Artificial Intelligence:** deduction systems (proof theory), expert systems, NLP, automated proof, multi-agent systems, etc.
- **Programming Languages:** logical programming, resolution functional programming, semantics, type systems, etc.
- **Software Engineering:** program verification, program design, specifications, etc.
- **Hardware:** circuit design verification, etc.

1.2 ROADMAP

We will be following a roadmap that is based on the following topics:

1. Propositional logic
 - Natural deduction, with which we will use it to reason about logical principles.
2. First Order Logic
3. Intuitionistic logic
 - $\underbrace{\vdash \varphi \iff \models \varphi}_{\leftarrow}$
 - λ calculus
 - Curry Howard
4. Modal Logics $\boxed{?}$, $\Diamond$

2 PROPOSITIONAL LOGIC

We are going to start with propositional logic, which is the simplest logical system.

2.1 SYNTAX

Propositional logic is composed by:

- Proposition symbols: $AT = \{p_0, p_1, p_2, \dots\} \cup \perp$ ($|\mathbb{N}| = \aleph_0$)
- Connectives:
 - Negation: $\neg\varphi$
 - Conjunction: $\varphi \wedge \psi$
 - Disjunction: $\varphi \vee \psi$
 - Implication: $\varphi \rightarrow \psi$
 - Bi-implication: $\varphi \leftrightarrow \psi$
- Auxiliary symbol: $(,)$

The set PROP of propositions is the *smallest* set X with the properties:

1. $p_i \in X (i \in \mathbb{N}), \perp \in X$
2. $\varphi, \psi \in X \Rightarrow (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\psi \leftrightarrow \varphi) \in X$
3. $\varphi \in X \Rightarrow (\neg\varphi) \in X$

We need the following induction principles in order to use proof by induction:

Theorem: Induction Principle

Let A be property when $A(\varphi)$ holds for all $\varphi \in \text{PROP}$ if:

1. $A(p_i)$ holds for all $i \in \mathbb{N}$ and $A(\perp)$ holds
2. $A(\varphi)$ and $A(\psi)$ hold implies $A(\varphi \wedge \psi), A(\varphi \vee \psi), A(\varphi \rightarrow \psi), A(\psi \leftrightarrow \varphi)$ hold
3. $A(\varphi)$ holds implies $A(\neg\varphi)$ holds

Theorem: Definition by Recursion

Let mappings $H_{\square} : A^2 \rightarrow A$ and $H_{\neg} : A \rightarrow A$ be given and let H_{at} be a mapping from the set of atoms into A , then there exists exactly one mapping $F : \text{PROP} \rightarrow A$ such that

Theorem: Primitive Recursion

$\exists! f : \mathbb{N} \rightarrow \mathbb{N}$ and $h : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ where $c \in \mathbb{N}$ such that $f(0) = c$ and $\forall n \in \mathbb{N}, f(n+1) = H(f(n))$:

- Ω class of satisfying sets Z
 - $(0, c) \in Z, (n, x) \in Z \Rightarrow (s(n), h(n, x))$
 - f intersection of all the sets above
 - f is a set and in Ω

We prove by induction that $\forall n \in \mathbb{N}, \exists! a \in \mathbb{N}$ such that $s \vdash (n, a) \in f$.

- **Base case:** $(0, c) \in f$ by definition. Suppose $\exists d \neq c$ and $f \setminus \{(0, d)\}$ is satisfied and properly contained in f .

Since I am violating the fact that f is the smallest one, this is impossible.

- **Inductive case:** Suppose $\exists! w(i, w) \in f$. By construction of f :

$$(s(i), h(i, w)) \in f$$

suppose $c \neq h(i, w)$ s.t. $(s(i), c) \in f$.

$$f \setminus \{(s(i), c)\} \text{ is satisfied and properly contained in } f$$

$$\Rightarrow \text{impossible}$$

2.2 SEMANTICS

The following two definitions are equal:

Definition: 1

A mapping $v : \text{PROP} \rightarrow \{0, 1\}$ is a valuation if:

- $v(p_i) \in \{0, 1\}$ for all $i \in \mathbb{N}$ and $v(\perp) = 0$
- $v(\varphi \wedge \psi) = \min(v(\varphi), v(\psi))$
- $v(\varphi \vee \psi) = \max(v(\varphi), v(\psi))$
- $v(\varphi \rightarrow \psi) = \max(1 - v(\varphi), v(\psi))$
- $v(\varphi \leftrightarrow \psi) = 1 \Leftrightarrow v(\varphi) = v(\psi)$
- $v(\neg\varphi) = 1 - v(\varphi)$

Definition: 2

A mapping $v : \text{PROP} \rightarrow \{0, 1\}$ is a valuation if:

- $v(p_i) \in \{0, 1\}$ for all $i \in \mathbb{N}$ and $v(\perp) = 0$
- $v(\varphi \wedge \psi) = 1$ iff $v(\varphi) = v(\psi) = 1$
- $v(\varphi \vee \psi) = 1$ iff $v(\varphi) = 1$ or $v(\psi) = 1$
- $v(\varphi \rightarrow \psi) = 0$ iff $v(\varphi) = 1$ and $v(\psi) = 0$
- $v(\varphi \leftrightarrow \psi) = 1$ iff $v(\varphi) = v(\psi)$
- $v(\neg\varphi) = 1$ iff $v(\varphi) = 0$